

Zishan Toufique

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SKILLS

- **Languages:** C, C++, Java, Python, HTML, CSS, JavaScript, My SQL
- **Technologies:** Python, C, C++, Java, Bash, Linux
- **Databases/Tools:** Nmap, Wireshark, Burp Suite, Kali Linux, Git, GitHub
- **Soft Skills:** Problem solving, Team collaboration, Time management, Adaptability

PROJECTS

Secure File Encryption & Decryption Tool | GitHub

Jun 2026 - Aug 2026

- Developed a desktop application for secure file encryption and decryption using AES-256-CBC, protecting sensitive files through password-based cryptographic operations.
- Implemented PBKDF2 key derivation, random salt generation, IV generation, and password validation to strengthen encryption security and prevent unauthorized file access.
- Designed a user-friendly Tkinter GUI for file selection, encryption/decryption operations, and secure handling of encrypted and decrypted files.
- Implemented secure file handling with separate encrypted and decrypted output locations, ensuring organized storage and reducing the risk of accidental exposure of sensitive files.
- **Tech Stack:** Python, Tkinter, PyCryptodome, AES-256-CBC, PBKDF2

Network Vulnerability Scanner | GitHub

Feb 2026 - May 2026

- Developed a Python-based network vulnerability assessment tool to identify active hosts, open ports, and potentially exposed network services.
- Integrated Nmap and socket programming to perform host discovery and port scanning, helping analyze the security posture of local networks.
- Implemented automated scan results to highlight accessible services and provide basic security recommendations for reducing unnecessary network exposure.
- Developed a structured scanning workflow to categorize detected ports and services, enabling faster identification of potentially risky network configurations and improving the overall assessment process.
- **Tech Stack:** Python, Nmap, Socket Programming, TCP/IP, Network Security, Linux

Smart Traffic Light System using Piezoelectric Plates and IR Sensors | GitHub

Oct 2025 - Jan 2026

- Developed an intelligent traffic control prototype that detects real-time vehicle presence using IR sensors and piezoelectric plates.
- Implemented dynamic signal control to prioritize lanes with waiting vehicles while avoiding unnecessary green signals for empty lanes.
- Designed the system to improve traffic flow, reduce idle waiting time, and provide an energy-conscious approach to automated traffic management.
- Integrated sensor-based inputs with the Arduino controller to automate traffic signal switching, enabling the system to respond dynamically to changing vehicle density across lanes.
- **Tech Stack:** Arduino, IR Sensors, Piezoelectric Sensors, Embedded C/C++, Circuit Design, Sensor Integration

CERTIFICATES

- Cyber Security 101 Certificate | [TryHackMe](#) Jul 2026
- Mastercard – Cybersecurity Job Simulation | [Forge](#) Jun 2026
- Introduction to Python | [Saylor](#) Feb 2026
- Networking Basics | [Cisco Networking Academy](#) Jan 2026

ACHIEVEMENTS

- Ranked among the **Top 4% of TryHackMe users**, demonstrating consistent hands-on practice in cybersecurity, networking, and ethical hacking.
- Completed multiple **cybersecurity and networking certifications**, including training in network security, ethical hacking, and red-team security practices.
- Developed hands-on cybersecurity projects involving **network vulnerability assessment, port scanning, encryption, and secure file handling**.

EDUCATION

- **Lovely Professional University** Phagwara, Punjab
• *Bachelor of Technology (Hons.) - Computer Science and Engineering; CGPA: 9.78* Aug 2025 - Present
- **Dhanbad Public School** Dhanbad, Jharkhand
• *Higher Secondary Education; Percentage: 91.0%* May 2022 - Mar 2024
- **De Nobili School** Dhanbad, Jharkhand
• *Secondary Education; Percentage: 80.0%* June 2021 - Mar 2022